Course milestone M7: First Reproducible Analysis (+ URC Abstract Internal Gate)

HONR 46400 · Evidence-Driven Research

What to submit

One document, `lastname_m07.pdf`. Everything this milestone asks for goes inside it, as sections in the order below. One artifact travels beside it, because it cannot live inside a PDF:

Submit it on Brightspace, under **Assignments** then **M07**. Brightspace carries the deadline.

This milestone presents **Book Milestone 7: Your first reproducible analysis** (checkpoint, version 1) of the course book, EDR|AI.

- `Shared, rerunnable Colab notebook link (or lastname_m07_first_analysis.ipynb)`: The milestone is graded partly by the instructor reopening the pipeline and rerunning it from a fresh runtime. A live, executable notebook with rerun permissions cannot exist inside a PDF, so the link or the .ipynb must accompany lastname_m07.pdf.

What this course adds

The required leakage audit

Before you finalize what you will claim, submit your verified draft to the **Prediction & Leakage Auditor**. This is the course's required AI reviewer role for this milestone; its full briefing is in `genai_studio/roles/prediction_leakage_auditor.md`. What it looks for depends on your route.

- **If your route predicts**, it hunts **data leakage**: a feature whose value is only settled at or after the outcome it is supposed to predict. Settle each flag with two checks inside your own pipeline. The **timing check** asks whether the feature's value is known before the prediction moment. The **correlation check** asks whether the feature tracks the outcome so tightly that it is nearly a copy of the answer. A feature that fails the timing check gets dropped or re-timed, however much it helps your score.
- **On every other route**, it audits your language instead. It hunts any sentence that quietly extends your result to units, times, or populations your design never reached. Bound or cut every sentence it flags.

For each flag, write your fix or your refutation, and settle it with a check in your own pipeline. The auditor can miss a real leak and invent a false one, so treat its flags as hypotheses to test, never as verdicts.

One more check on code you delegated: the **known-answer test**. Run a piece of your code on a tiny input whose right answer you already know. For example, feed your group-difference function two three-row groups you can average by hand, and confirm it returns exactly that. Keep the test in the notebook, next to the code it verifies.

Your conference abstract, and its internal gate

The course also asks for the abstract you would submit to the Undergraduate Research Conference. Write roughly 150 to 250 words on your project as it now stands: the question, what you measure, the analysis your pipeline executes, the verified result with its uncertainty, and the boundary around the claim.

This is an **internal gate**. Bring the draft to the Friday studio, which runs as an abstract clinic, and clear it with me there before it goes out externally. Its clock belongs to the conference rather than to your research schedule, which is why it lands in this week's work.

The gate has one rule that does not bend. Your abstract may use only the claims your current verified result licenses. An abstract does not clear if it promises a causal finding your route does not license, reaches a population your data never touched, or reports as settled a number no stress test has yet touched. Your result is verified this week and not yet attacked. The robustness work ahead can narrow it, so word the abstract to survive that without becoming false.

A notebook that can be rerun

Your pipeline notebook is seeded with `SEED = 464`, and it loads data from the source your M6 governance record documents. Beside your PDF, share the Colab notebook (or hand in the `.ipynb`) with sharing set so the instructor can open **and rerun** it. Before you submit, confirm that a run from a fresh runtime, top to bottom, prints the numbers your write-up reports.

How this is judged

The Book Milestone rubric governs your PDF. Two course rules sit on top of it. An abstract that has not cleared the gate is unfinished work. A missing AI Research Ledger row scores Craft 0 and the submission is returned to you for completion before it is graded.

Milestone 7: Your first reproducible analysis

Open the milestone workbook in Colab

This chapter closes Studio 7: Produce a reproducible first analysis. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A route-specific result with an uncertainty statement, a restart-and-run-all verification, an environment record, and a claim-to-output check.

What you bring

You also carry forward the last milestone's artifact: **Milestone 6: Your data and measurement, governed**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 22: AI as Programmer: your first computed number, its delegated code read line by line, and the sentence naming what the number does not cover.
- Lesson 23: AI as Analytical Assistant: your headline estimate with its uncertainty statement and cycle log, two independently re-derived numbers, a placebo run where licensed, the clean-restart record, the environment record, and the claim-to-output trace.

The practice

1. Write the analysis you declared, and only that analysis, before looking at anything else.
2. Produce the result with its uncertainty statement, in the form your Contract specified.
3. Restart and run everything from a clean state; if the numbers move, the pipeline is the finding.
4. Record your environment, and check that every claim you plan to make traces to a specific output.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** What you send to a tool during analysis is a disclosure decision each time.
- **Evidence, provenance, and reproducibility.** Every number in your result traces to a data cell and a line of code.
- **AI activity, verification, and human decisions.** Delegate the writing of code freely; verify every returned number against the data yourself.
- **Uncertainty, claim boundary, and revision history.** A result reported without uncertainty is not yet a result.

Where this milestone sits

You arrived carrying Milestone 6: Your data and measurement, governed. Milestone 8 exists to attack this result; the cleaner your record here, the more the attack can teach you. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Failing to reproduce your own analysis sends you back here before any stress-testing is meaningful.

How this milestone is assessed

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: A route-specific result with an uncertainty statement, a restart-and-run-all verification, an environment record, and a claim-to-output check	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

> ! Blocking gate
>
> No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.